

# Syllabus

Spring semester 2026

|              |                      |             |              |                 |         |   |                       |           |
|--------------|----------------------|-------------|--------------|-----------------|---------|---|-----------------------|-----------|
| Course Title | General Biology<br>1 | Course Code | CB3201       |                 | Credit  | 4 | Classroom             | Chang 408 |
|              |                      | Type        | AP, Elective |                 | Hours   | 4 |                       |           |
| Teacher      | Name                 | Class no.   | Office       | Office hours    | Contact |   | E-mail                |           |
|              | Ko, In-Jeong         |             | Changjo 403  | Mon5,Wed6, Fri5 |         |   | koin2@ksa.kaist.ac.kr |           |

## 1. Course Description

| Course Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <ul style="list-style-type: none"> <li>■ Learn AP biology at a college level.</li> <li>■ Learn and integrate the basic biological structure and function concepts to understand more complex biological systems.</li> <li>■ Familiarize yourself with English expressions used in biology.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
| <ol style="list-style-type: none"> <li>1. Biology and Its Themes</li> <li>2. Atoms and Molecules</li> <li>3. The Chemistry of Water</li> <li>4. Carbon: The Basis of Molecular Diversity</li> <li>5. Biological Macromolecules and Lipids</li> <li>6. Energy and Life</li> <li>7. Cell Structure and Function</li> <li>8. Cell Membrane</li> <li>9. Cellular Signaling</li> <li>10. Cell Respiration</li> <li>11. Photosynthesis Processes</li> <li>12. Mitosis</li> <li>13. Sexual Life Cycles and Meiosis</li> <li>14. Mendelian Genetics</li> <li>15. Linkage and Chromosomes</li> <li>16. Nucleic Acids and Inheritance</li> <li>17. Expression of Genes</li> <li>18. Control of Gene Expression</li> </ol> |  |

## 2. Core Target Competencies

| Subject Core Competencies                                                                                                                                                                                                                                                                                                           |                                  |                       |               |               |                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------------|---------------|---------------|--------------------------------------------------------------|
| Basic concept comprehension                                                                                                                                                                                                                                                                                                         | Analytical and creative thinking | Problem-solving skill | Communication | Collaboration | Habits and mindset (self-efficacy, resilience, perseverance) |
| Students learn AP-level course content through lectures and individual/group studies. Focus on cultivating basic comprehension of biology concepts, scientific thinking, and problem-solving capabilities through group problem-solving and presentation projects during class and developing research and collaboration abilities. |                                  |                       |               |               |                                                              |

### 3. Textbooks and References

- A. Textbook: Biology, A Global Approach (2021) Campbell et al., 12th Edition, Pearson
- B. References: Problem Solving Workbook (Provided)
- C. Resources for student presentation  
Find your topic based on the materials from 'RCSB Protein Data Bank' and HHMI. A student presents the topic (10~15 min). Further details about the oral presentation will be announced separately.

### 4. Class Method

- A. Lecture
- B. Students' group activity: problem solving & questioning
- C. Students' presentation for further study

### 5. Evaluation

#### A. Evaluation Criteria

| Workbook1 | Quiz & Homework2 | Student Presentation3 | Exam4   |       | Attendance & Attitude5 | Total |
|-----------|------------------|-----------------------|---------|-------|------------------------|-------|
|           |                  |                       | Midterm | Final |                        |       |
| 10%       | 10%              | 10%                   | 30%     | 30%   | 10%                    | 100%  |

#### B. Evaluation Method

- (1) **Workbook** (10): Before class, please prepare for the problems in the workbook. You may be asked to participate in the class discussion activities.
- (2) **Quiz & Homework** (10): Quiz 1(3), Quiz 2(3), Homework 1(2), Homework 2(2)
- (3) **Student Presentation** (10):
  - Find your topic based on the materials from 'RCSB Protein Data Bank' and HHMI. Each student presents their topic (10-15 min). More detailed info will be given in the Google Classroom.
  - Presentation preparation: You must submit the draft PPT one week before your presentation date to get the teacher's feedback. For that, you upload your file to GC or email it to your teacher; you must include your ID number and name in the file name. (For example, Presentation 3\_24-OOO\_김DD) If not, 0.5 penalty points for a misnamed file.
- (4) **Midterm & Final exam** (60): Short answer and/or essay test, 'fill in the blank' in the figure/table, and multiple choice test will be given.
- (5) **Attendance & Attitude** (10): Points will be deducted for each absence, lateness, or sleeping.(1pt for each absence with no report, 0.5 pts for lateness, 0.2 pts penalty for any sleeping student during each class)

### 6. Criteria for Academic Honors and Course Credit

- ☞ Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.

- To earn course credit, students must attend at least three-fourths (3/4) of the total class hours.
- Method for calculating make-up scores for excused absences:  
For exams missed due to an official excused absence or illness, the recognized score will be calculated by taking the percentile rank of the student's total scores from other exams in the same subject and applying it to the equivalent percentile score of the missed exam, then multiplying by the attendance weighting factor (100% for official excused absences; 80% for absences due to illness).

## 7. Weekly Plans

| Week (period)              | Topic and Contents                                                                                                                                                                                       | (Notes)                                                                    |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Week 1</b><br>2/23-2/27 | Introduction to the course<br>1. Biology and Its Themes<br>2. Atoms and Molecules<br>3. The Chemistry of Water<br>4. Carbon: The Basis of Molecular Diversity<br>5. Biological Macromolecules and Lipids | -Opening Day/<br>Entrance Ceremony<br>(2/23, Mon)<br>-Self-study Ch 1 to 4 |
| Core Competencies          | ♦ concept comprehension, scientific thinking ability, problem solving ability, self-efficacy                                                                                                             |                                                                            |
| <b>Week 2</b><br>3/2-3/6   | 6. Energy and Life                                                                                                                                                                                       | 3/2(Mon) Substitute holiday<br><br>- Plan your presentation                |
| Core Competencies          | ♦ concept comprehension, scientific thinking ability, problem solving ability, collaboration ability                                                                                                     |                                                                            |
| <b>Week 3</b><br>3/9-3/13  | 7. Cell Structure and Function                                                                                                                                                                           | - Amino acids (quiz)<br>- <b>Student Presentation</b>                      |
| Core Competencies          | ♦ concept comprehension, scientific thinking ability, problem solving ability, collaboration ability                                                                                                     |                                                                            |
| <b>Week 4</b><br>3/16-3/20 | 8. Cell Membrane                                                                                                                                                                                         | - <b>Student Presentation</b>                                              |
| Core Competencies          | ♦ concept comprehension, scientific thinking ability, problem solving ability, collaboration ability                                                                                                     |                                                                            |
| <b>Week 5</b><br>3/23-3/27 | 9. Cellular Signaling                                                                                                                                                                                    | - <b>Student Presentation</b>                                              |
| Core Competencies          | ♦ concept comprehension, scientific thinking ability, problem solving ability, collaboration ability                                                                                                     |                                                                            |
| <b>Week 6</b><br>3/30-4/3  | 10. Cell Respiration                                                                                                                                                                                     | - <b>Student Presentation</b>                                              |
| Core Competencies          | ♦ concept comprehension, scientific thinking ability, problem solving ability, collaboration ability                                                                                                     |                                                                            |
| <b>Week 7</b><br>4/6-4/10  | 11. Photosynthesis Processes                                                                                                                                                                             | - Glycolysis, Krebs cycle, Calvin cycle (quiz)                             |
| Core Competencies          | ♦ concept comprehension, scientific thinking ability, problem solving ability, collaboration ability                                                                                                     |                                                                            |
| <b>Week 8</b><br>4/13-4/17 | Midterm Exams (4/14-4/17)                                                                                                                                                                                |                                                                            |

| Week (period)               | Topic and Contents                                                                                   | (Notes)                                                            |
|-----------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Week 9</b><br>4/20-4/24  | 12. Mitosis                                                                                          |                                                                    |
| Core Competencies           | ♦ concept comprehension, scientific thinking ability, problem solving ability, collaboration ability |                                                                    |
| <b>Week 10</b><br>4/27-5/1  | 13. Sexual Life Cycles and Meiosis                                                                   | 5/8(Fri)~9 SAF<br>- <b>Student Presentation</b>                    |
| Core Competencies           | ♦ concept comprehension, scientific thinking ability, problem solving ability, collaboration ability |                                                                    |
| <b>Week 11</b><br>5/4-5/8   | 14. Mendelian Genetics                                                                               | Ch14. Genetics problem (Homework)<br>- <b>Student Presentation</b> |
| Core Competencies           | ♦ concept comprehension, scientific thinking ability, problem solving ability, collaboration ability |                                                                    |
| <b>Week 12</b><br>5/11-5/15 | 15. Linkage and Chromosomes                                                                          | Ch15. Genetics problem (Homework)<br>- <b>Student Presentation</b> |
| Core Competencies           | ♦ concept comprehension, scientific thinking ability, problem solving ability, collaboration ability |                                                                    |
| <b>Week 13</b><br>5/18-5/22 | 16. Nucleic Acids and Inheritance                                                                    | - <b>Student Presentation</b>                                      |
| Core Competencies           | ♦ concept comprehension, scientific thinking ability, problem solving ability, collaboration ability |                                                                    |
| <b>Week 14</b><br>5/25-5/29 | 17. Expression of Genes                                                                              | - <b>Student Presentation</b>                                      |
| Core Competencies           | ♦ concept comprehension, scientific thinking ability, problem solving ability, collaboration ability |                                                                    |
| <b>Week 15</b><br>6/1-6/5   | 18. Control of Gene Expression                                                                       |                                                                    |
| Core Competencies           | ♦ concept comprehension, scientific thinking ability, problem solving ability, collaboration ability |                                                                    |
| <b>Week 16</b><br>6/8-6/12  | Make-up class week                                                                                   |                                                                    |
| <b>Week 17</b><br>6/15-6/19 | Final Exams                                                                                          |                                                                    |

※ The above plans may change according to the academic schedule.